

Data Modeling (Part 1)

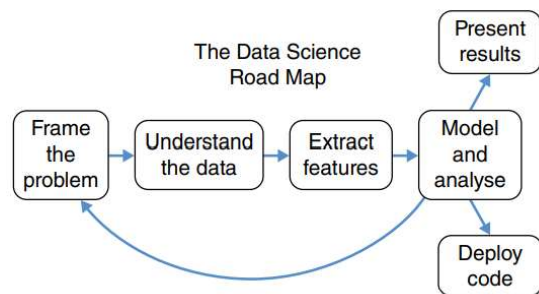
Le Ngoc Thanh
lnthanh@fit.hcmus.edu.vn
Department of Computer Science

Ho Chi Minh City

Contents

- Data science and machine learning
- Machine learning architecture
- Regression model

Process



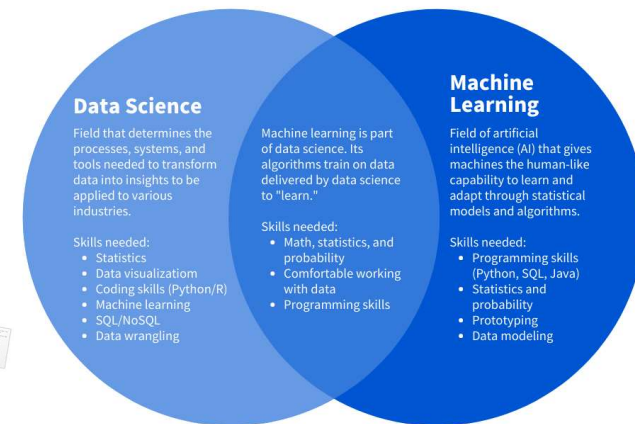
After preprocessing

ID	NAME	SALARY	COUNTRY	CITY	ACTIONS
1	Dakota Rice	\$36,730	Niger	Quet Lambout	♥ 🔍 ✕
2	Minerva Hooper	\$23,789	Comoros	Smea West	♥ 🔍 ✕
3	Sage Rodriguez	\$56,142	Netherlands	Batavia	♥ 🔍 ✕
4	Philip Chaney	\$18,735	Korea, South	Overland Park	♥ 🔍 ✕
5	Doris Greene	\$63,542	Malawi	Feldkirchen in Kärnten	♥ 🔍 ✕
6	Mason Porter	\$78,415	Chile	Gloucester	♥ 🔍 ✕
7	Alden Chen	\$63,529	Finland	Gary	♥ 🔍 ✕
8	Colton Hodges	\$93,901	Nicaragua	Daft	♥ 🔍 ✕
9	Colton Hodges	\$93,901	Nicaragua	Daft	♥ 🔍 ✕
10	Colton Hodges	\$93,901	Nicaragua	Daft	♥ 🔍 ✕
11	Colton Hodges	\$93,901	Nicaragua	Daft	♥ 🔍 ✕
12	Colton Hodges	\$93,901	Nicaragua	Daft	♥ 🔍 ✕

Data Science Process

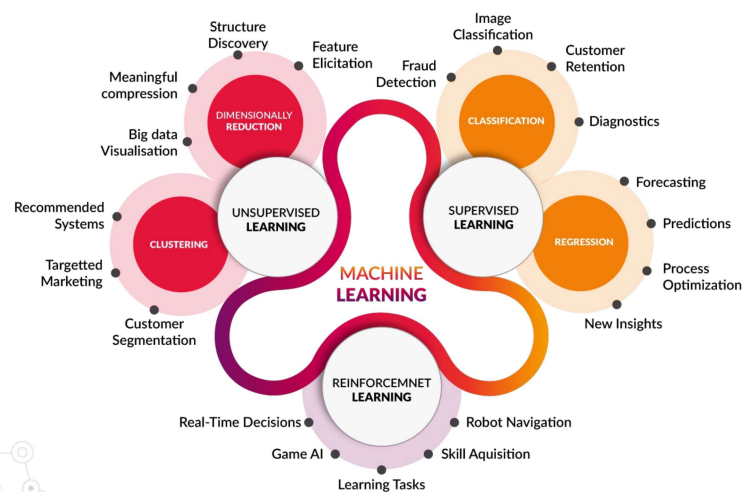
- Give the question to answer
- Collecting data
- Data Discovery & preprocessing to obtain data that can be analyzed
- Data analysis** (in visualizations, statistics, machine learning)
→ answers (hypotheses) for the question
- Evaluation
- Decision Making

Data Science vs. Machine Learning



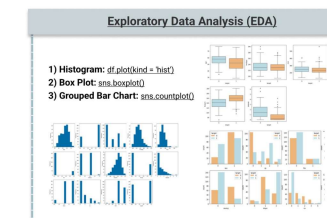
<https://www.coursera.org/articles/data-science-vs-machine-learning>

ML Tasks

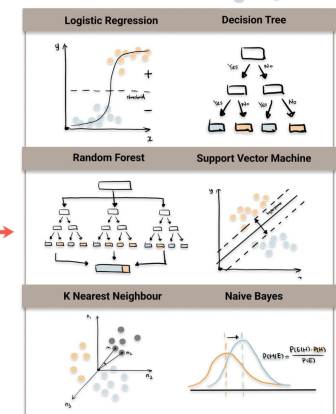


Machine Learning Choice

- Before implementing the machine learning (ML) model, the data scientist needs to **identify (several) branches** in ML that can solve the given problem.



Visualization and Statistics

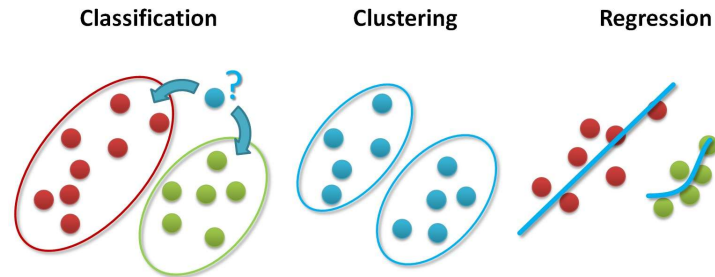


Machine Learning

The course's focus

◎ In this course, we focus on **three main groups** of ML:

- Regression
- Classification
- Clustering

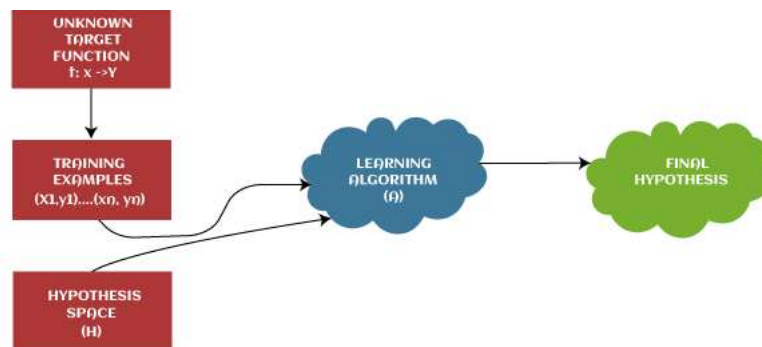


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- ◎ Data science and machine learning
- ◎ **Machine learning architecture**
- ◎ Regression model

After hypothesis

◎ The job of a learning algorithm to **find the best suitable hypothesis** for a problem.



After hypothesis

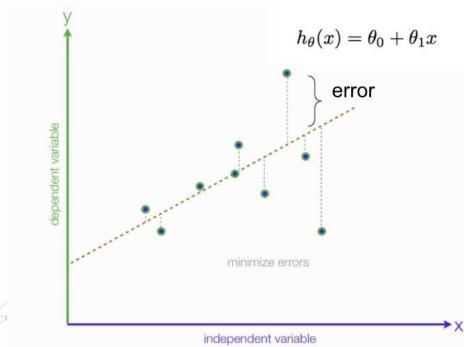
◎ To choose the suitable hypothesis, we need to **define the loss function**.

$$\mathcal{L}(y - \hat{y}) = \sum_{i=1}^n (y - \hat{y}_i)^2$$

Machine learning = *iterative procedure to find a minimum of loss for the given data.*

After loss function design

- ⦿ We are looking for **what parameters to produce the lowest loss rate** for given dataset, so we need **the process to optimize the function (fitting)**.



Hypothesis:

$$h_{\theta}(x) = \theta_0 + \theta_1 x$$

Parameters:

$$\theta_0, \theta_1$$

Cost Function:

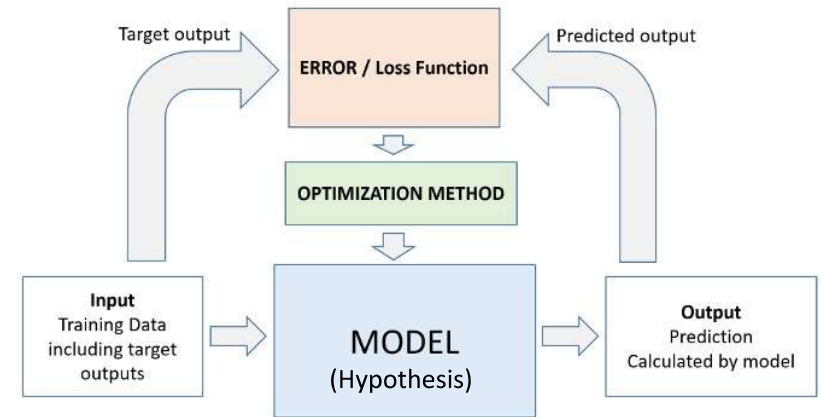
$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2$$

Goal:

$$\underset{\theta_0, \theta_1}{\text{minimize}} J(\theta_0, \theta_1)$$

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General model learning architecture



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- ⦿ Data science and machine learning
- ⦿ Machine learning architecture
- ⦿ **Regression model**
 - Linear regression
 - Non-linear regression
 - Over- and Under-Determined Systems
 - Model selection
 - Overfitting

Regression

- ⦿ Consider a set of n data points:

$$(x_1, y_1), (x_2, y_2), (x_3, y_3), \dots, (x_n, y_n)$$

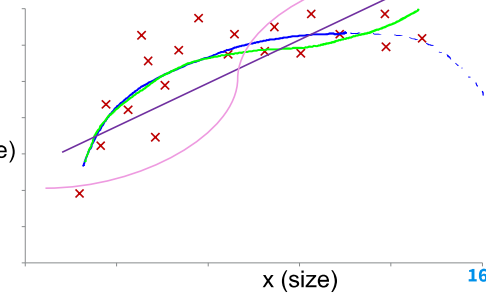
- ⦿ Purpose:

- Select a function $f(\cdot)$ and fit it to the data (**curve fitting = regression**)

$$Y = f(A, \beta)$$

Size in feet ² (x)	Price (\$) in 1000's (y)
100	10
800	150
1534	315
852	178

y
(price)



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Linear regression

- Assume that a **line** is fitted through the points (**hypothesis**)

$$f(x) = \beta_1 x + \beta_2$$

- The loss function is **MSE** (mean-squares error)

$$E(f) = \frac{1}{n} \sum_{k=1}^n (f(x_k) - y_k)^2 = \frac{1}{n} \sum_{k=1}^n (\beta_1 x_k + \beta_2 - y_k)^2$$

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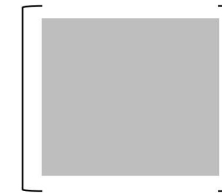
Linear regression

- The optimization method: **derivatives**
- Generalization, the 2×2 system:

$$\mathbf{Ax} = \mathbf{b}$$

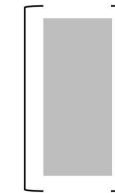
Model terms

$\mathbf{A}$



Loadings

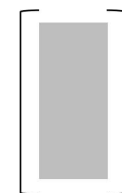
$\mathbf{x}$



=

Outcomes

$\mathbf{b}$



=

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 - Fit Function**
 - Gradient descent
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Nonlinear regression

- How with nonlinear regression? For example:

$$f(x) = \beta_2 \exp(\beta_1 x)$$

- The MSE function:

$$E(\beta_1, \beta_2) = \sum_{k=1}^n (\beta_2 \exp(\beta_1 x_k) - y_k)^2$$

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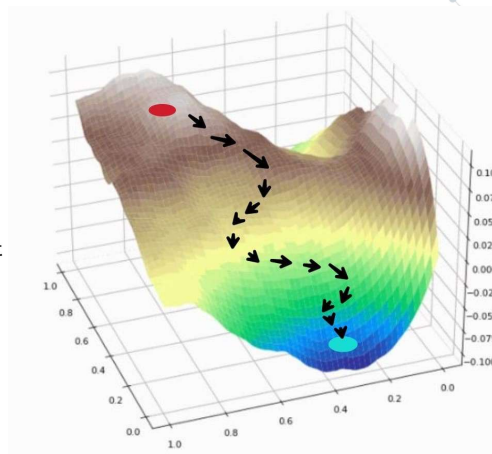
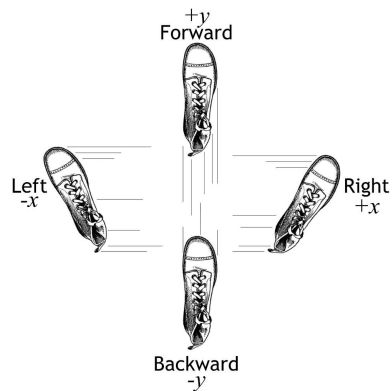
Go to downhill



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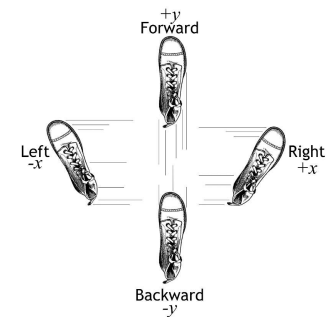
Go to downhill



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Go to downhill

- What means if direction vector is:
 $[x, y]$
= [which way is down in x direction, which way is down in y direction]
 $[-1, 1]$
- To actually move downhill, we move to:
 $\Rightarrow [x_{new}, y_{new}] = [x, y] + [-1, 1]$



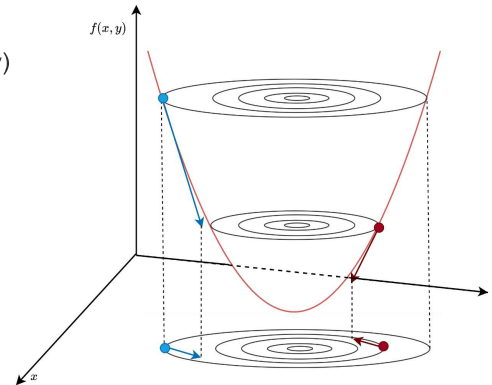
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Go to downhill

- Generally, to move in xy space toward the minimum point, we need identify:

- Moving direction (increase/decrease x and y)
- Rate of change (based on slope)

⇒ It is a **direction vector**



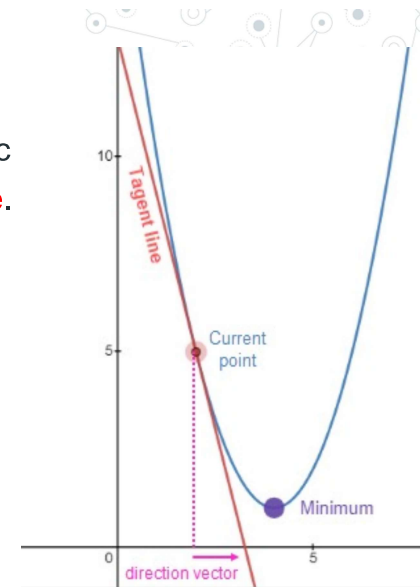
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Direction vector

- The **derivative** of a function at a specific point gives the **slope of the tangent line**.

$$f'(x) = \lim_{(x_1 - x_0) \rightarrow 0} \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$

- Why is the **tangent line** considered as a **direction vector**?



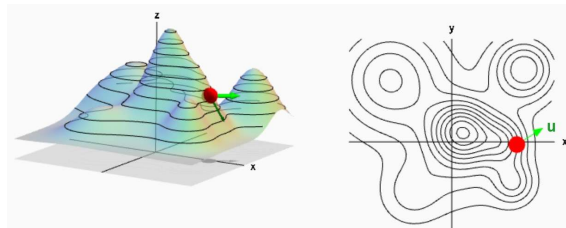
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Directional derivative

- If you stand at some point $\mathbf{a} = (x_0, y_0)$, the **slope of the ground in front of you** will **depend on the direction you are facing**.
- To **calculate the slope in any direction**, we **derivative in this direction**.
⇒ called the **directional derivative**.

$$D_{\mathbf{u}}f(x_0, y_0)$$

where $\mathbf{u} = (u_1, u_2)$ is an **unit vector** that points in the direction in which we want to compute the slope.



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Gradient

- The **gradient** of f at any point tells you:
 - a **direction** is the **steepest** from that point with respect to the x, y plane
 - how **steep** it is (the slope of the hill in that direction)

$$\nabla f(x, y) = \begin{bmatrix} \frac{\partial f(x, y)}{\partial x} \\ \frac{\partial f(x, y)}{\partial y} \end{bmatrix} = \frac{\partial f(x, y)}{\partial x} \hat{\mathbf{x}} + \frac{\partial f(x, y)}{\partial y} \hat{\mathbf{y}}$$

- The partial derivatives give the **slope in the positive x direction** and the **slope in the positive y direction**.

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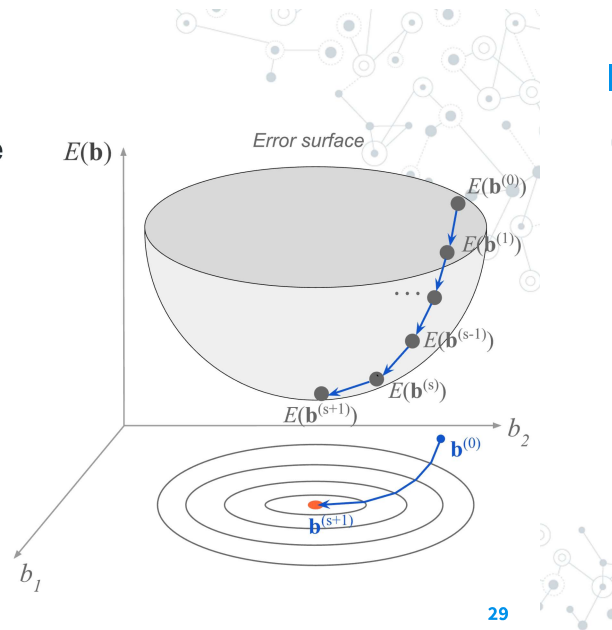
Gradient Descent

- As we **update**, we want the value of $f(x, y)$ to **decrease**.

- When it stops decreasing, (x_0, y_0) will have arrived at the position giving the minimum value of $f(x, y)$.

- The next position at time step t :

$$\mathbf{x}_{t+1} = \mathbf{x}_t - \nabla f(\mathbf{x}_t)$$

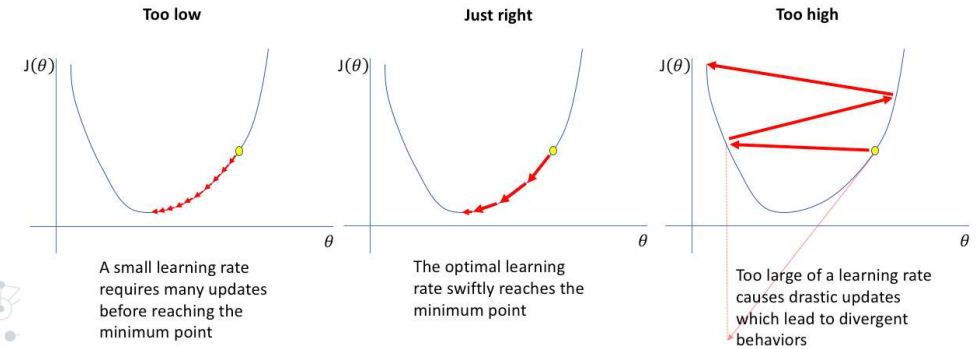


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Issues: Learning rate

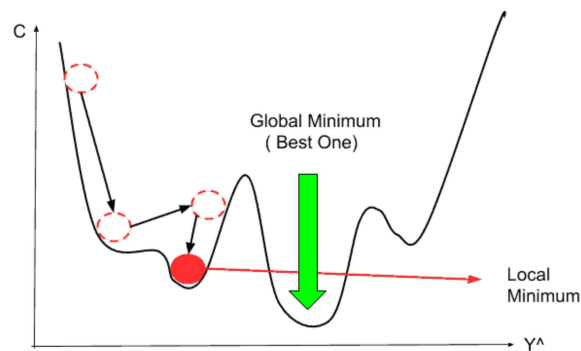
- Need to restrict the size of the steps by shrinking the direction vector using a **learning rate** η , whose value is less than 1:

$$\mathbf{x}_{t+1} = \mathbf{x}_t - \eta \nabla f(\mathbf{x}_t)$$



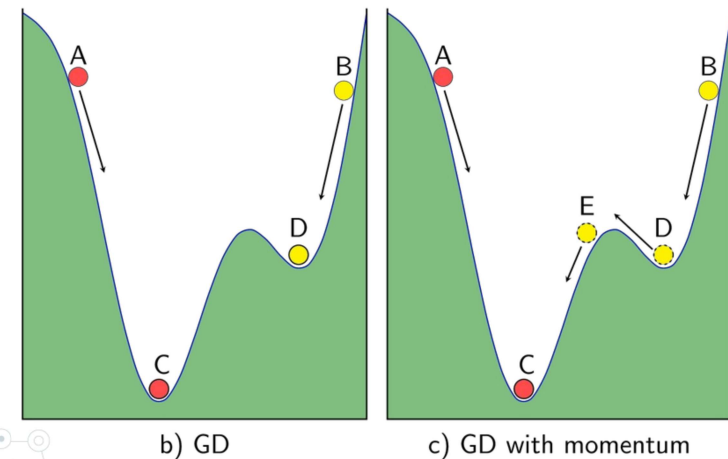
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Issues: Starting point (non-linear function)



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Momentum



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Summary for nonlinear regression

◎ The nonlinear optimization procedure:

- The **initial guess**
- **Step size** η
- **Computing the gradient** efficiently

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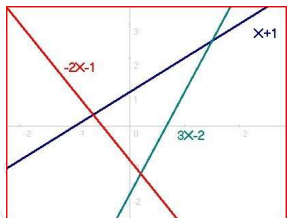
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Over-determined systems

◎ **Over-determined systems** have more constraints (equations) than unknown variables.

- **No solutions** satisfying the linear system.
- **Approximate solutions** to minimize a given error.



$$\begin{array}{ccc} \text{Model terms} & \text{Loadings} & \text{Outcomes} \\ A & x & b \\ \left[\begin{array}{c} \text{ } \\ \text{ } \\ \text{ } \end{array} \right] & \left[\begin{array}{c} \text{ } \\ \text{ } \\ \text{ } \end{array} \right] & = \left[\begin{array}{c} \text{ } \\ \text{ } \\ \text{ } \end{array} \right] \end{array}$$

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Under-Determined Systems

◎ **Under-determined systems** have more unknowns than constraints.

- **an infinite number of solutions.**
- **some choice of constraint must be made.**

$$\begin{array}{ccc} \text{Model terms} & \text{Loadings} & \text{Outcomes} \\ A & x & b \\ \left[\begin{array}{c} \text{ } \\ \text{ } \end{array} \right] & \left[\begin{array}{c} \text{ } \\ \text{ } \\ \text{ } \end{array} \right] & = \left[\begin{array}{c} \text{ } \\ \text{ } \end{array} \right] \end{array}$$

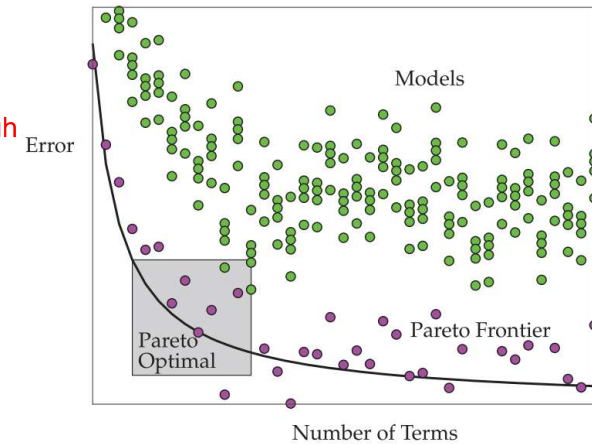
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Model Selection

- ◎ Model selection is not simply about reducing error, it is about producing a model that has a **high degree of interpretability, generalization and predictive capabilities.**

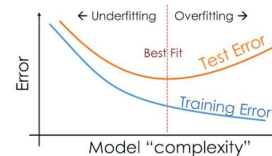
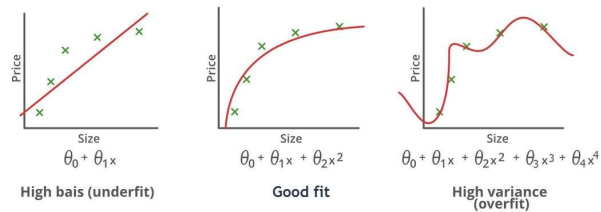


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Overfitting

- ◎ The production is too closely to a particular set of data, and may therefore fail to fit to predict future observations reliably.
 - Overfitting does not allow for generalization.



The End

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